

STAT230A Lecture 18 Leave-One-Out Formula (Application 2)

Logic ▾

对于 leave-one-out formula

$$\hat{\beta}_{[-i]} = \hat{\beta} - \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i \hat{\varepsilon}_i$$

其可被用于:

- Gauss Update Formula
- Outlier Detection

1 Leave-One-Out Formula 的应用: Outlier Detection

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对于 Normal linear model, 我们要求 $Y = X\beta + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$

但有时 observations 中有一部分满足该 assumption, 而另一部分并不满足, 这些 observations 被称为 out-of-distribution 或 outlier points, 我们希望能发现这些 observations

对于 Normal linear model, 有 $\hat{\varepsilon} \sim \mathcal{N}(\vec{0}, \sigma^2(I - H))$, 其可被近似为 $\mathcal{N}(\vec{0}, \hat{\sigma}^2(I - H))$, 因此一种想法是: check $\hat{\varepsilon}$ 或与 $\hat{\varepsilon}$ 相关的量是否服从特定分布

1.1 Definition: Standardized Residual

定义 standardized residual 为:

$$\text{stand } r_i = \frac{\hat{\varepsilon}_i}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}}$$

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注意到对于 stand r_i , 其分子分母 correlated ($\hat{\sigma} = \frac{\hat{\varepsilon}^\top \hat{\varepsilon}}{n-p}$), 因此较难求出其 asymptotic distribution

1.2 Definition: Studentized Residual

定义 studentized residual 为:

$$\text{stud } r_i = \frac{\hat{\varepsilon}_{[-i]}}{\sqrt{\frac{\hat{\sigma}_{[-i]}^2}{1 - h_{ii}}}}$$

其中:

- $\hat{\varepsilon}_{[-i]} = y_i - x_i^\top \hat{\beta}_{[-i]}$ 为 predicted residual
- $\hat{\sigma}_{[-i]}^2 = \frac{(\hat{\varepsilon}^{(-i)})^\top \hat{\varepsilon}^{(-i)}}{(n-1)-p}$ 为去除 (X_i, y_i) 后得到的 variance estimator
- $\hat{\varepsilon}^{(-i)} = Y_{[-i]} - X_{[-i]} \hat{\beta}_{[-i]}$

且对于 Normal Linear Model, 有

$$\text{stud } r_i \sim t_{n-1-p}$$

⚠ Remark: Outlier Detection ▾

基于 studentized residuals, 我们可以将它们逐个与 t_{n-1-p} 比较并进行 hypothesis testing, 但需要注意这是一个 multiple hypothesis testing problem

1.3 Cook's Distance

🔗 Logic ▾

通过 testing studentized residuals, 我们可以得知哪些 points 偏离了 NLM 的假设, 但我们并不知道这些 points 多大程度上影响了 $\hat{\beta}$ 的估计

我们可以考虑使用 leave-one-out formula, 但这会得到一个 vector of difference (得到的是 $\hat{\beta}_{[-i]} - \hat{\beta}$), 我们希望得到一个数来概括 change to $\hat{\beta}$

由此我们可以考虑使用 Cook's distance, 其可被视作 standardized distance between $\hat{\beta}_{[-i]}$ 和 $\hat{\beta}$

第 i 个 observation 的 Cook's distance 被定义为:

$$\text{Cook}_i = \frac{(\hat{\beta}_{[-i]} - \hat{\beta})^\top (X^\top X)(\hat{\beta}_{[-i]} - \hat{\beta})}{p\hat{\sigma}^2}$$

1.4 Theorem: Cook's Distance 与 Standardized Residual 的联系

Cook's Distance 与 Standardized Residual 存在以下联系:

$$\text{Cook}_i = \text{stand } r_i^2 \cdot \frac{h_{ii}}{p(1 - h_{ii})}$$

⚠ Remark: ▾

该公式说明 Cook's distance 同时由两部分决定:

- $\text{stand } r_i^2$: 衡量该点在 **response direction** 上是否异常
- $\frac{h_{ii}}{1 - h_{ii}}$: 衡量该点是否具有 **high leverage**

因此, 一个 observation 要有较大的 Cook's distance, 通常需要同时满足:

- 较大的 residual
- 较大的 leverage

🔗 Proof ▾

由 leave-one-out formula, 有

$$\hat{\beta}_{[-i]} - \hat{\beta} = -\frac{1}{1 - h_{ii}}(X^\top X)^{-1}x_i\hat{\varepsilon}_i$$

因此,

$$\begin{aligned}
\text{Cook}_i &= \frac{(\hat{\beta}_{[-i]} - \hat{\beta})^\top (X^\top X)(\hat{\beta}_{[-i]} - \hat{\beta})}{p\hat{\sigma}^2} \\
&= \frac{1}{p\hat{\sigma}^2} \left(\frac{1}{1 - h_{ii}} \hat{\varepsilon}_i, x_i^\top (X^\top X)^{-1} \right) (X^\top X) \left(\frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i \hat{\varepsilon}_i \right) \\
&= \frac{\hat{\varepsilon}_i^2}{p\hat{\sigma}^2(1 - h_{ii})^2} x_i^\top (X^\top X)^{-1} x_i \\
&= \frac{\hat{\varepsilon}_i^2}{p\hat{\sigma}^2(1 - h_{ii})^2} h_{ii}
\end{aligned}$$

由于 standardized residual 被定义为

$$\text{stand } r_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma}\sqrt{1 - h_{ii}}},$$

因此

$$\text{Cook}_i = (\text{stand } r_i)^2 \cdot \frac{h_{ii}}{p(1 - h_{ii})}.$$